

Naive Bayes & LDA

Fast, interpretable generative models for multi-class classification problems.

1 Project Overview

I implemented Gaussian Naive Bayes and Linear Discriminant Analysis (LDA) to classify Jira bug tickets and silicon defects. Both algorithms effectively modeled class distributions. I achieved accuracies of 87.50% and 85.70%, showcasing their power in practical settings.

Naive Bayes: Uses Bayes' theorem under the independence assumption.

Gaussian Naive Bayes: Assumes features follow a Gaussian distribution for each class.

LDA: Finds feature combinations that maximize class separation.

2 Datasets

2.1 Jira Bug Ticket Routing

Property	Value
Description	Contains bug tickets routed to six engineering teams.
Total Rows	8,000
Columns	9 (8 features + 1 target)
Features	Ticket type, severity, project, etc.
Class Balance	Balanced classes across six categories.
Train / Test Split	80/20

2.2 Silicon Defect Classification

Property	Value
Description	Classifies defect types from wafer electrical measurements.
Total Rows	6,000
Columns	8
Features	Electrical measurements, defect indicators.
Class Balance	Balanced defect types and clean die.
Train / Test Split	80/20

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Gaussian Naive Bayes	Chosen for its simplicity and effectiveness in classification.
Multinomial Naive Bayes	Utilized for multi-class classification scenarios.
LDA	Selected for its ability to reduce dimensionality while preserving class separability.

I loaded and explored both datasets, followed by training Gaussian Naive Bayes and LDA models. Lastly, I evaluated model performance using confusion matrices and accuracy metrics.

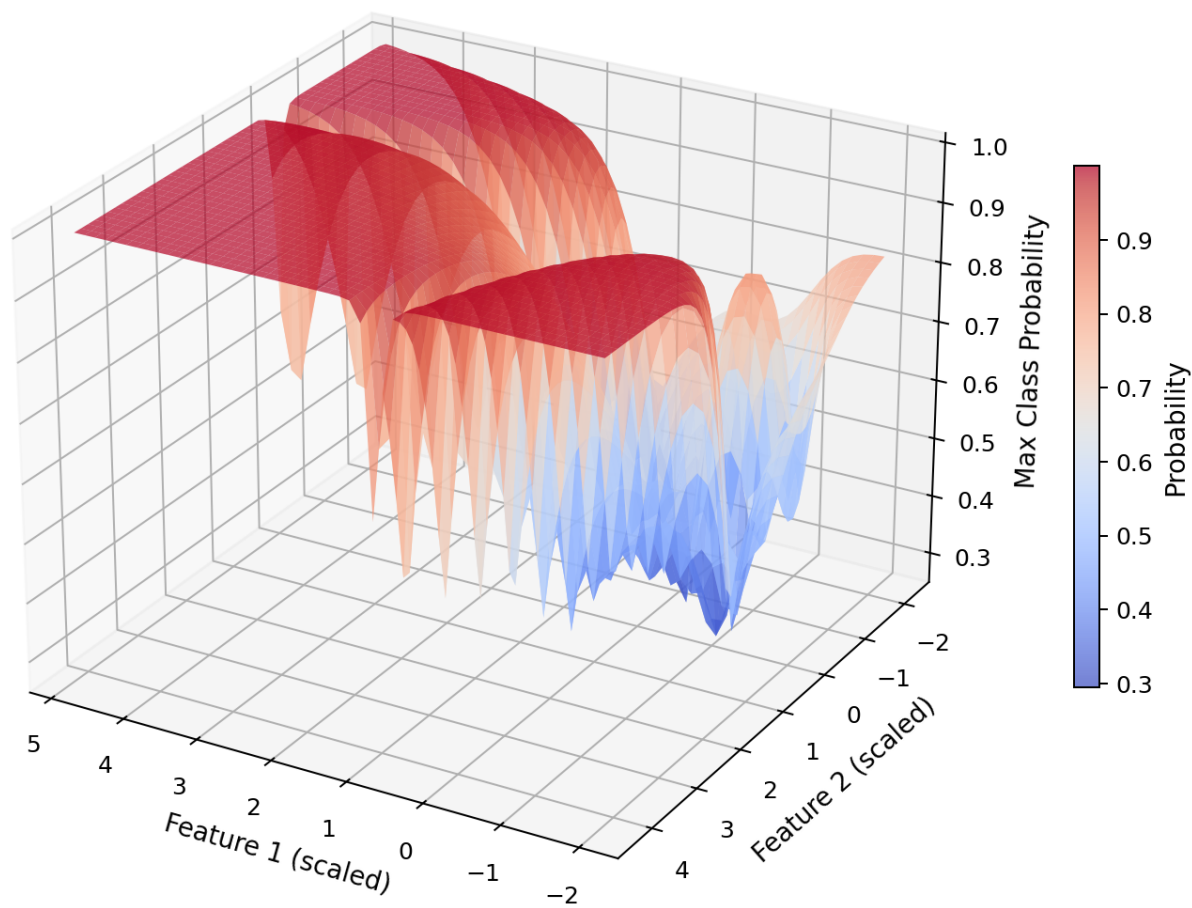
4 Results

LDA slightly outperformed Gaussian Naive Bayes in accuracy. The results indicate that LDA is more effective in capturing the relationships among features for these datasets, improving classification outcomes.

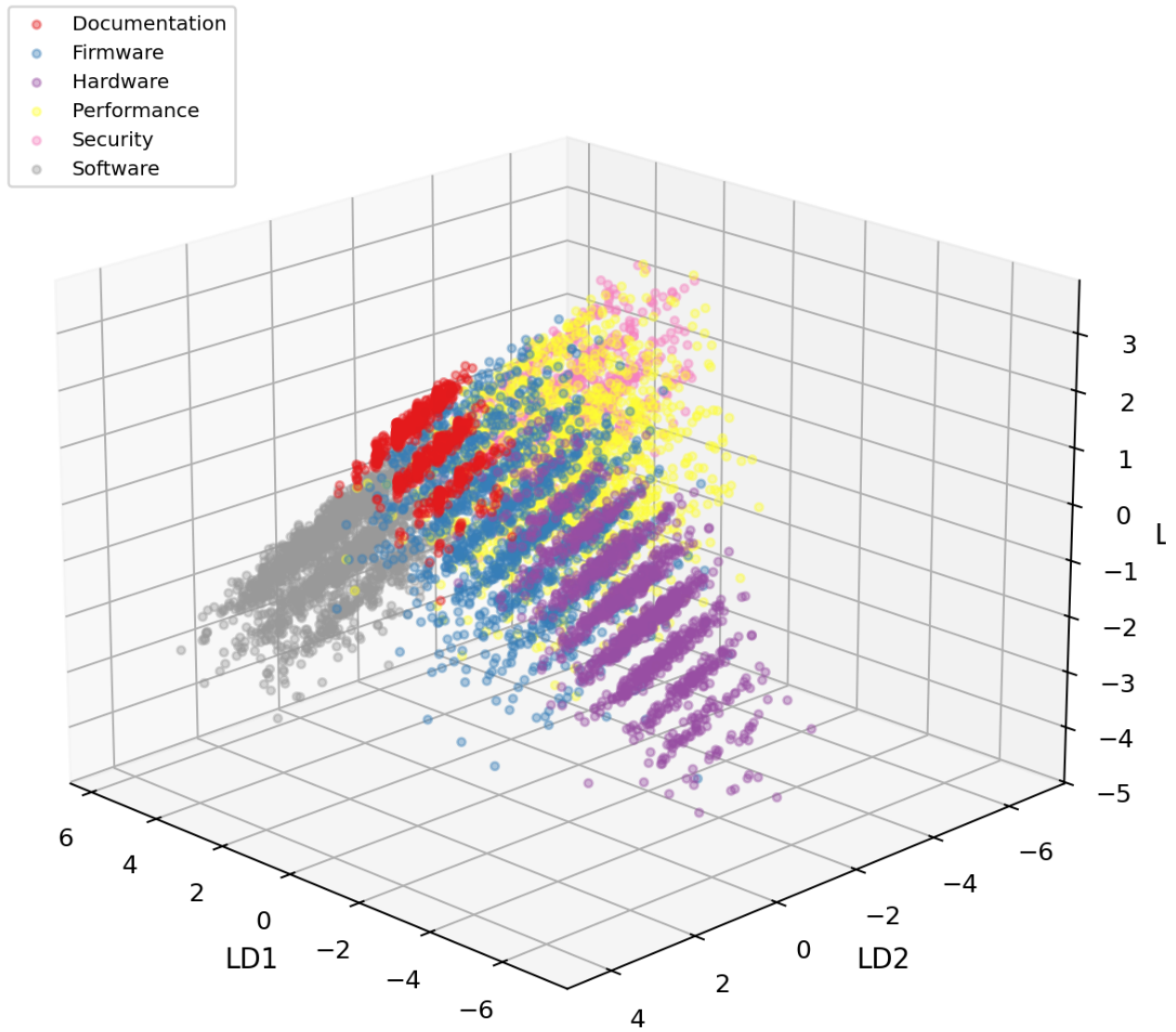
Metric	Value
Accuracy	0.8750
Accuracy	0.8570

Best Result **Accuracy: 0.8750**

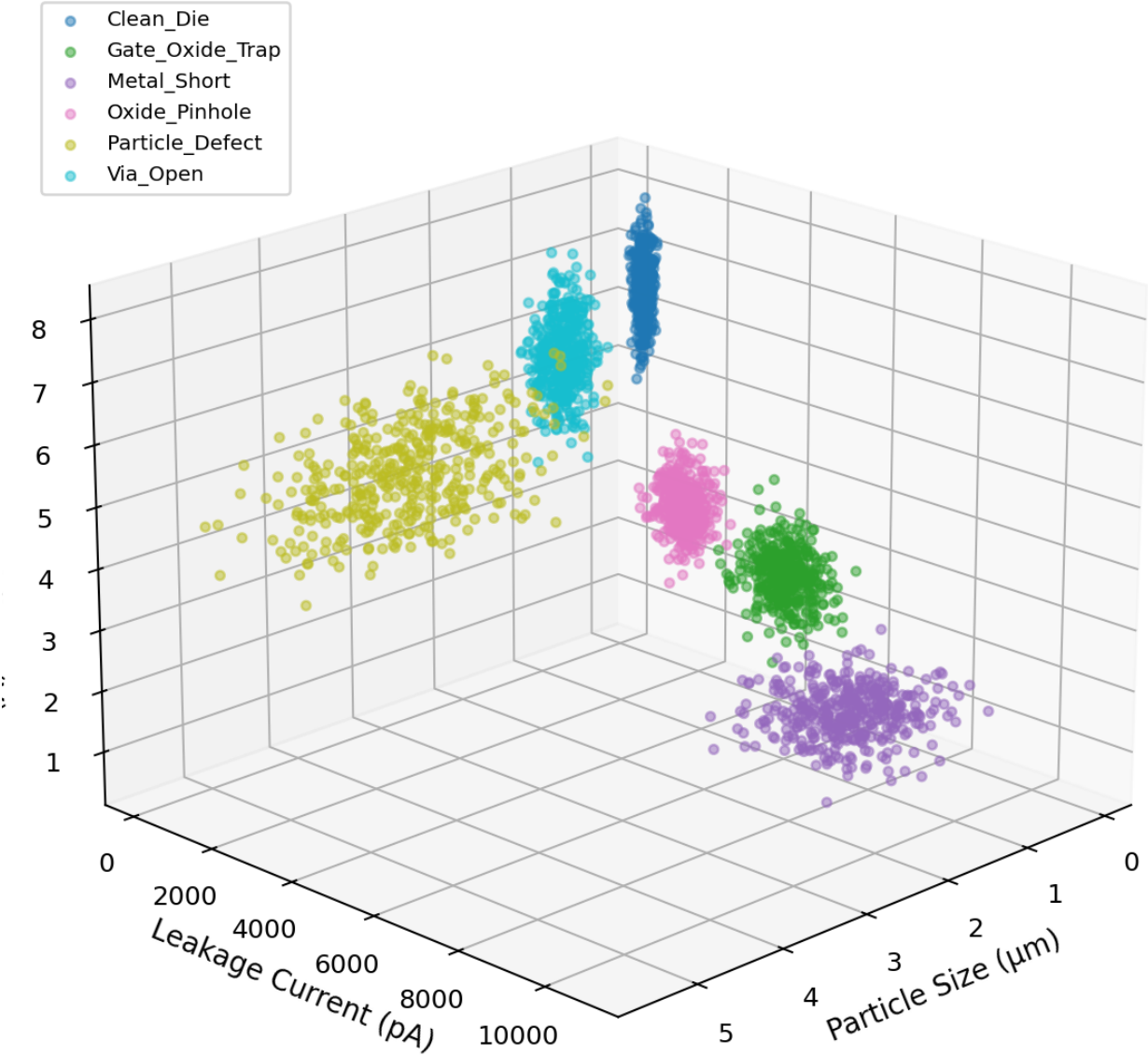
Naive Bayes Posterior Probability Surface
JIRA Bug Routing

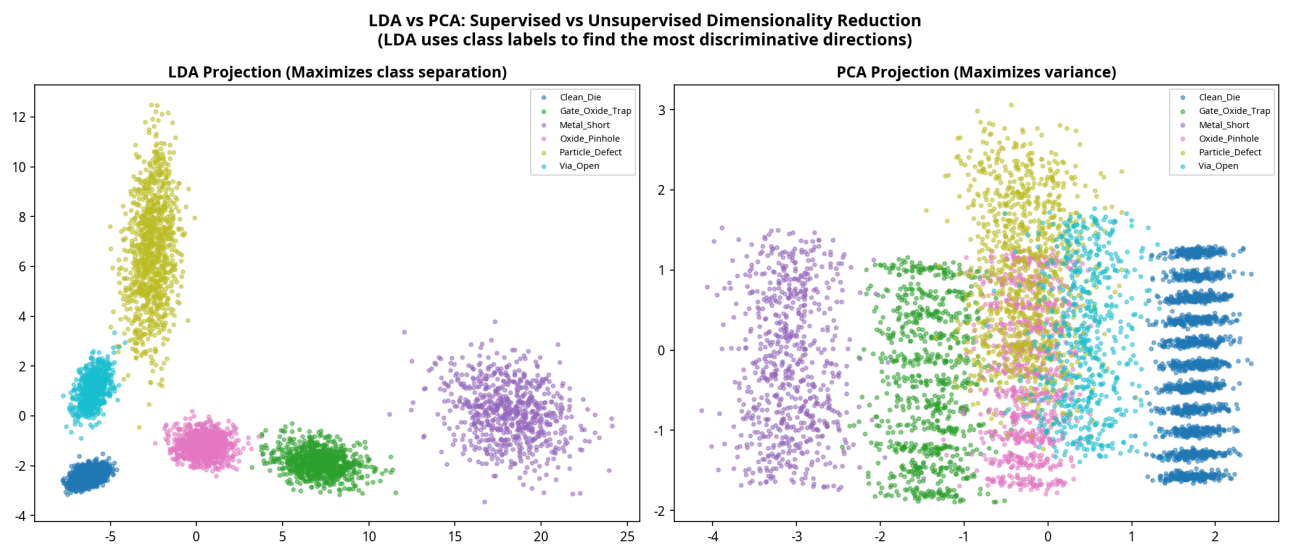
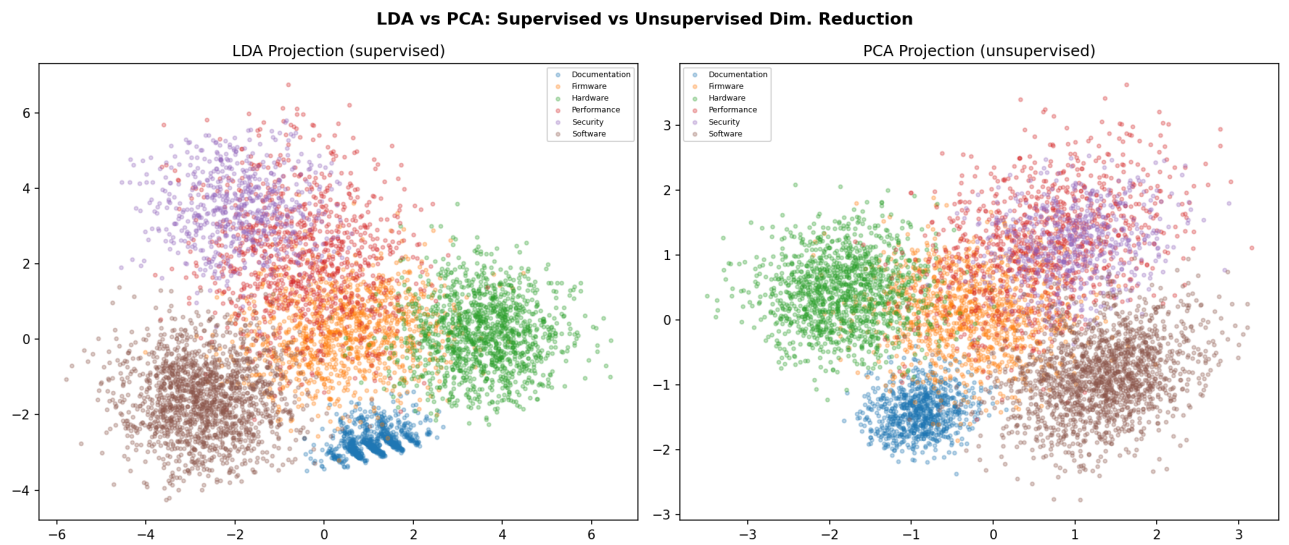
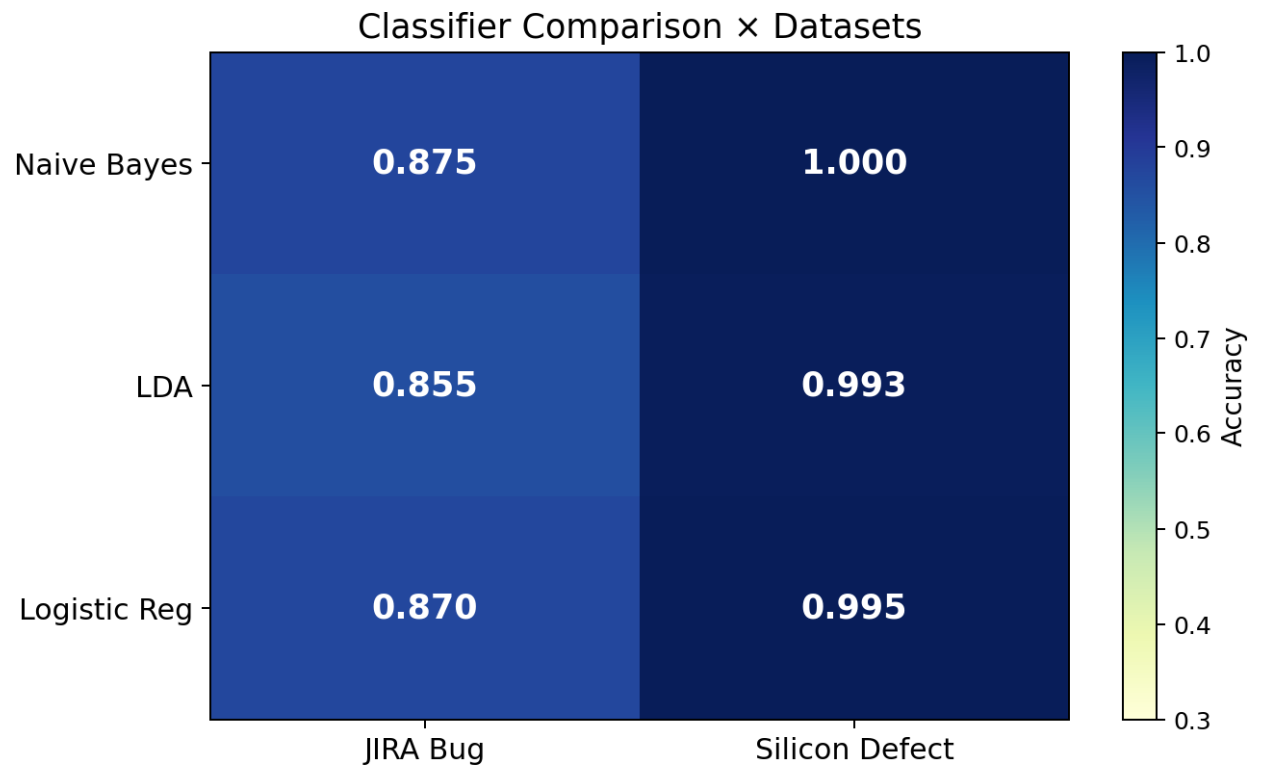


LDA 3D Projection — JIRA Bug Categories

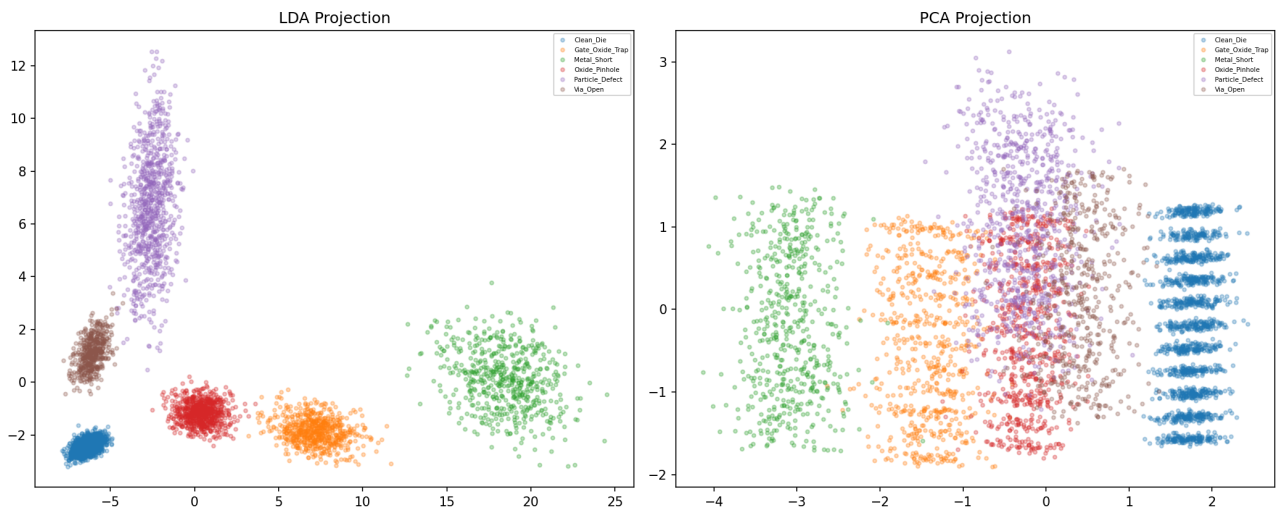
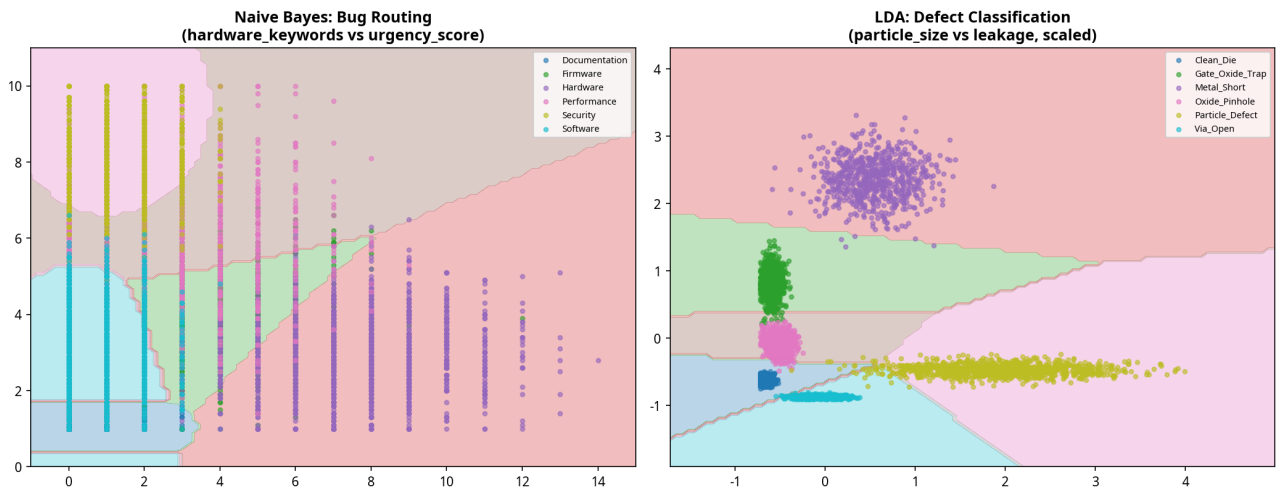
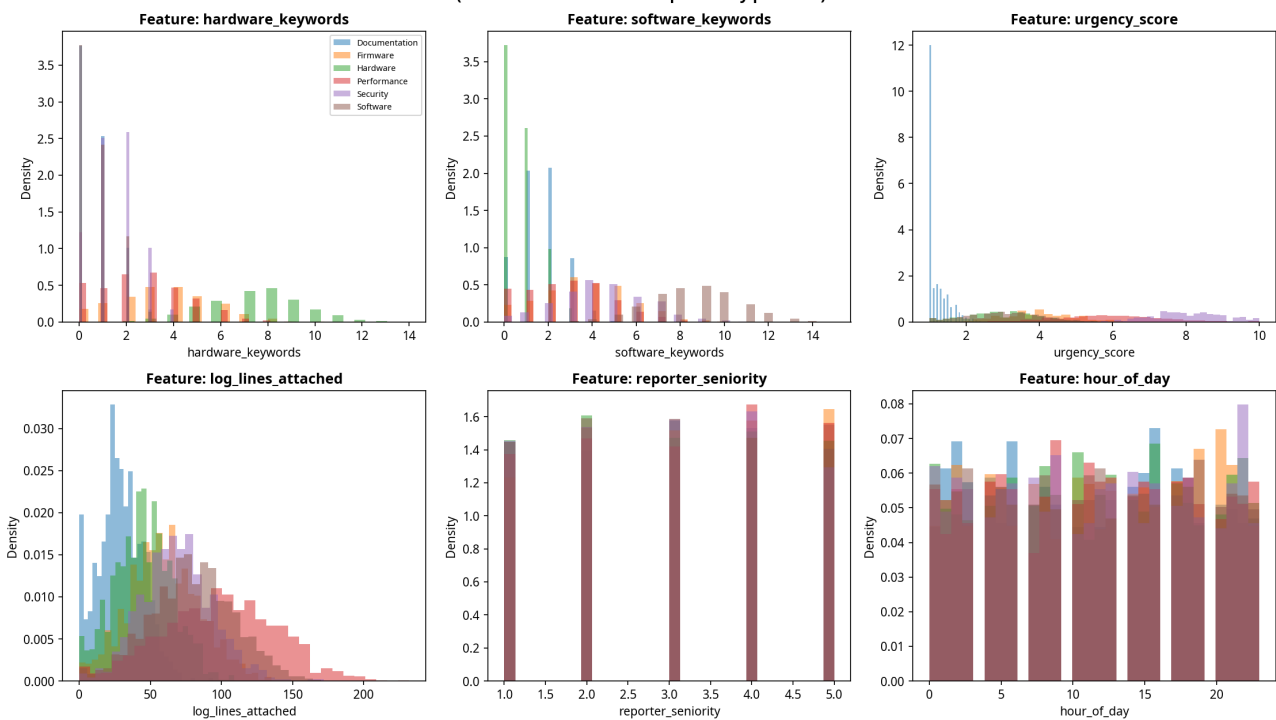


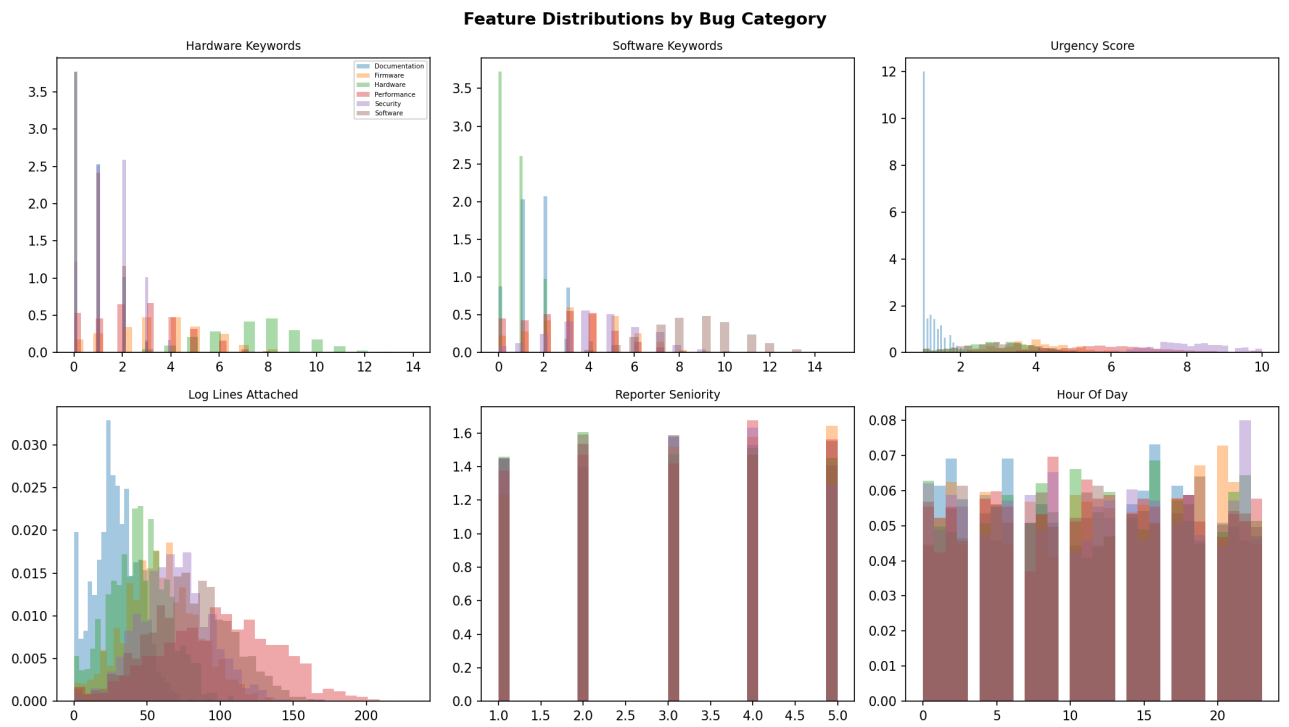
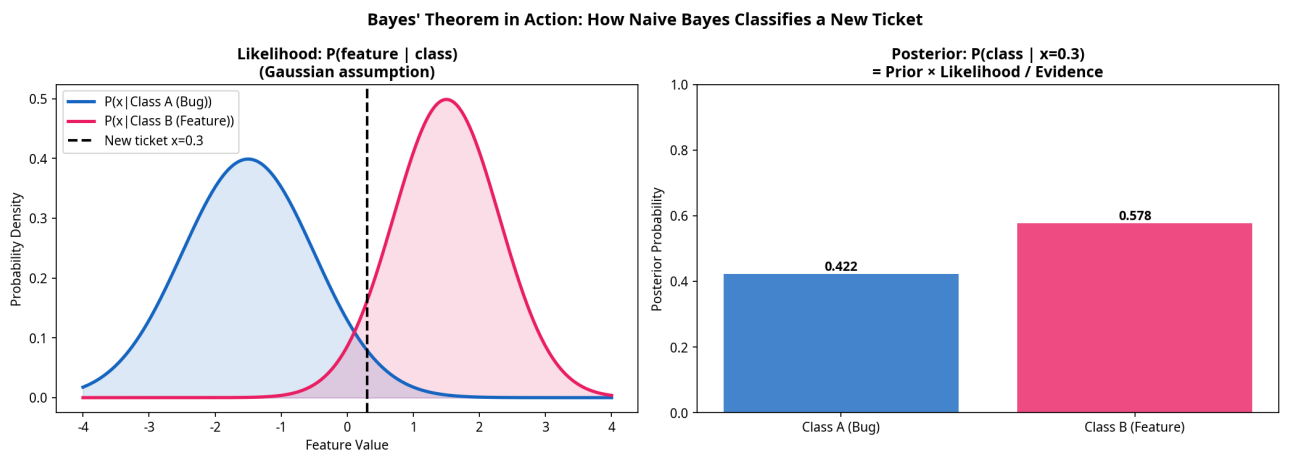
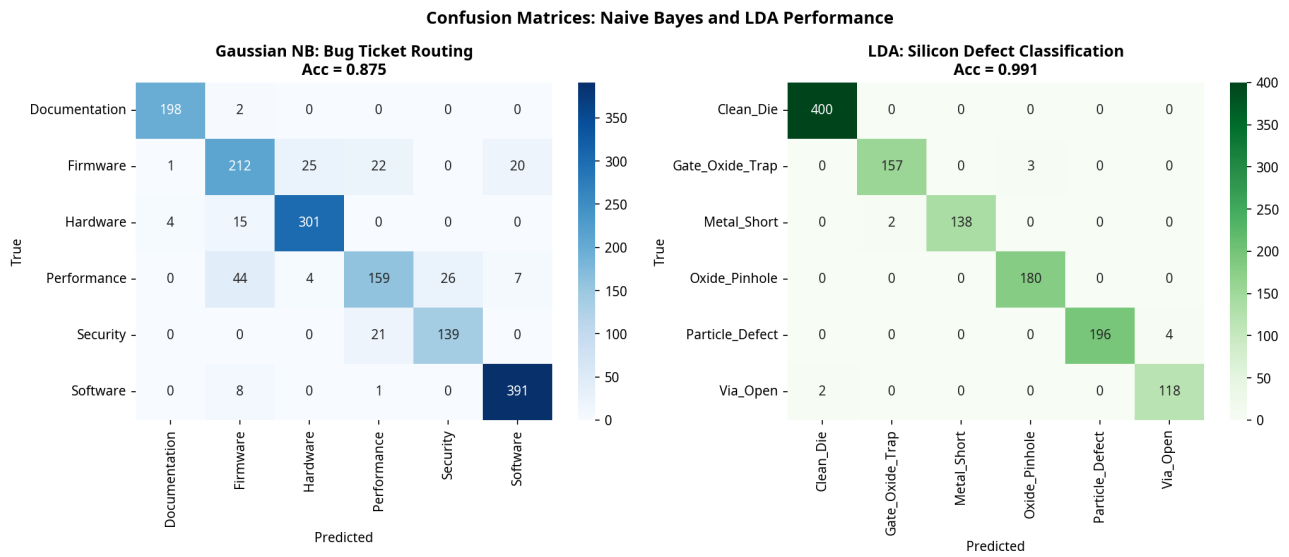
3D Silicon Defect Feature Space
Color = Defect Type

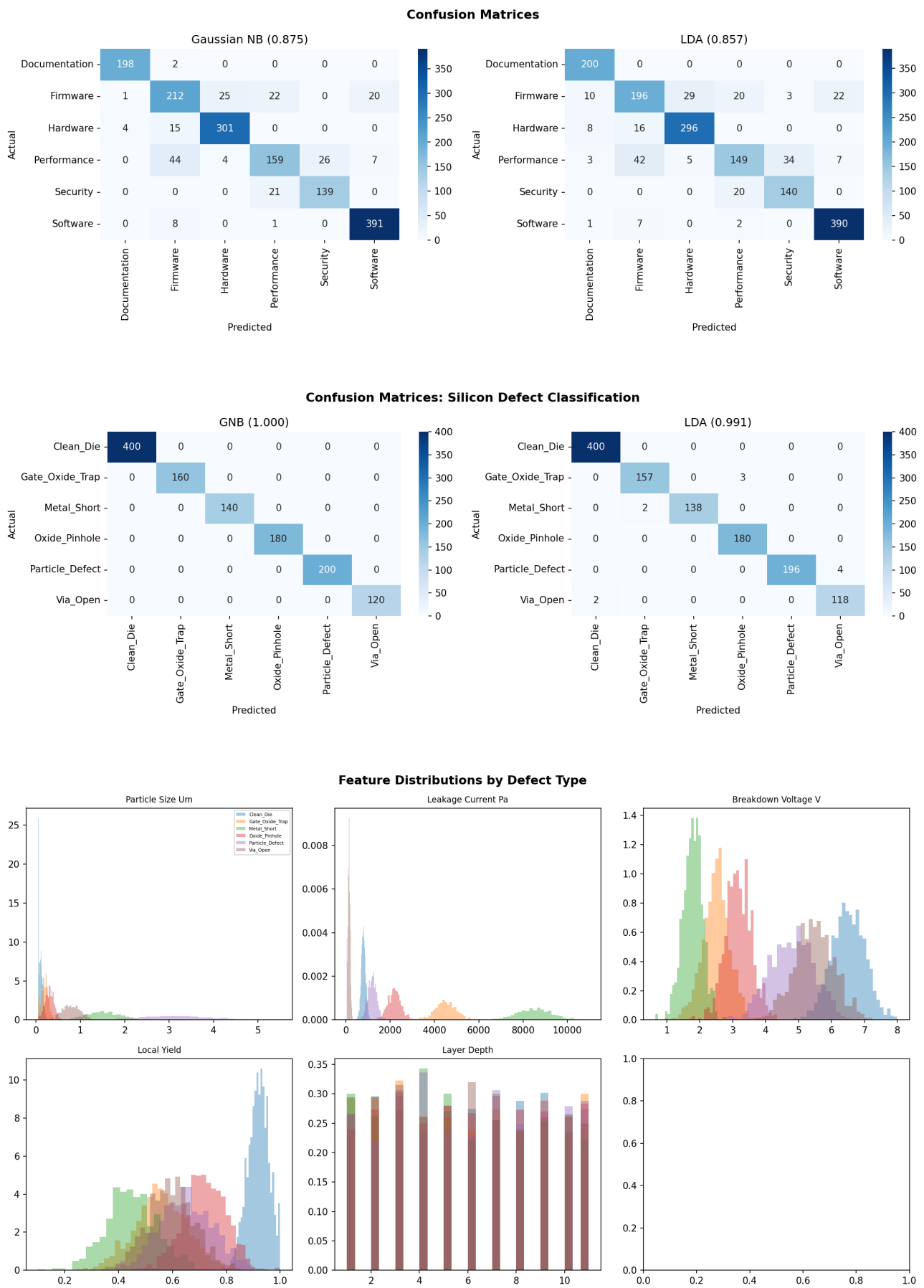




LDA vs PCA: Silicon Defect Classes

Decision Boundaries: Naive Bayes vs LDA
(Both are linear/quadratic — fast and interpretable)Naive Bayes: Class-Conditional Distributions
(Each feature modeled independently per class)





Info	Detail
GitHub	https://github.com/AIML-Engineering-Lab/008_naive_bayes_lda
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

- Naive Bayes performs well with independent features.
- LDA shines in cases requiring dimensionality reduction.
- Confusion matrices provide clear model performance insight.